

# Red Hat Research

Bringing great research ideas into open source communities

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Name(s)

Title

Red Hat Research



# Welcome slide (the call to action)



# What we'll be discussing today

Insert agenda items here

And here...

.....

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# Red Hat Research: Subheading for section

# Red Hat Research: Leveraging research partnerships to propel emerging technologies

Red Hat Research connects Red Hat engineers with professors, researchers, and students to bring great research ideas into open source communities. Our activities around the world have produced grants from government and industry, papers at top conferences, and results that have landed in open source projects of all kinds. Red Hat Research welcomes participation from research-minded individuals around the world.

## Research Areas



AI/ML



Cloud/DS



Hardware and  
the OS



Security, Privacy,  
Cryptography



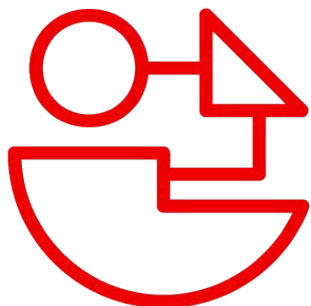
Testing and Ops

# Red Hat Research: AI/ML

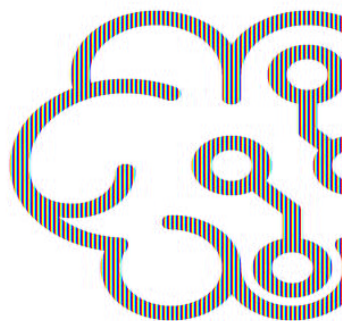


The combination of academic approaches and industry use cases promises is accelerating the adoption of AI/ML beyond supercomputing labs and theoretical research. Our research spans the gamut of algorithm development, data hygiene, AI methods, and attacking old challenges with AI approaches.

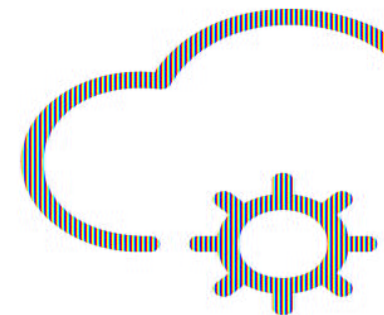
## Featured Projects



Tuning the Linux kernel



Disinformation detection at scale



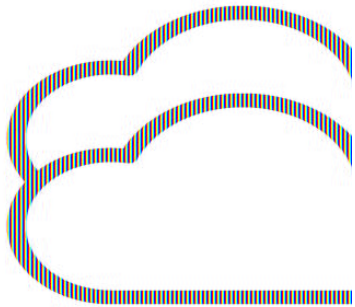
AI for CloudOps

# Red Hat Research: Cloud/DS

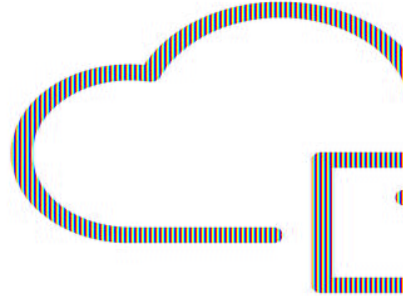


For a user, the cloud is a streamlined experience hiding an extreme degree of complexity, but engineers there is still much work to be done. Networking, infrastructure, scalability, self-awareness, self-healing, and automation to keep it manageable and secure are just some of the problems that our projects look to address, and their “lab” is very successful public-cloud alternatives that host other scientific and academic research projects.

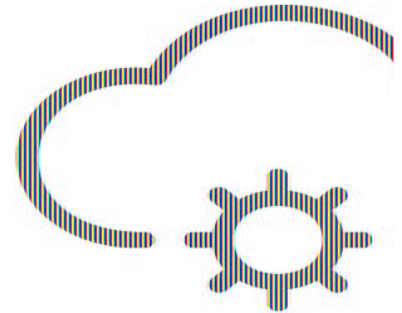
## Featured Projects



Insert project name



Insert project name



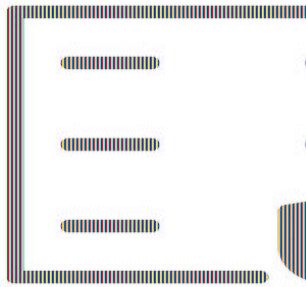
Insert project name

# Red Hat Research: Hardware and the OS



The success of all computing, even the cloud, relies on the interdependencies between hardware features and software OS that exposes them to applications. Edge computing and the performance and scale requirements of cloud are driving the evolution of new hardware types, and software innovation must keep pace. We actively seek projects that amplify hardware innovation and push software-defined capabilities forward.

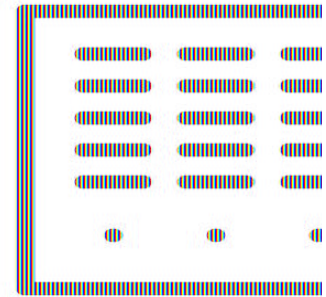
## Featured Projects



Insert project name



Insert project name



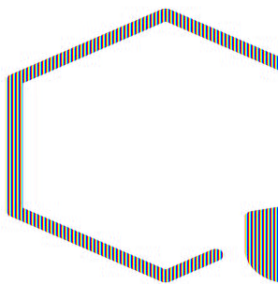
Insert project name

# Red Hat Research: Security, privacy, and cryptography

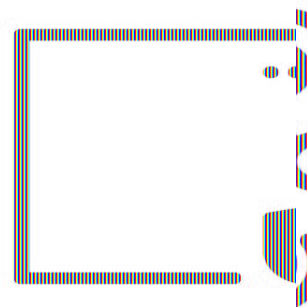


Enterprise IT is not structured to have the time and money to invest in far-reaching, forward-looking security solutions. Research in these areas thrives but does not easily find its way to industry. An open source approach allows researchers to share their work immediately and have it reviewed by thousands of security experts, so innovations in cryptography, privacy and cybersecurity have immediate impact.

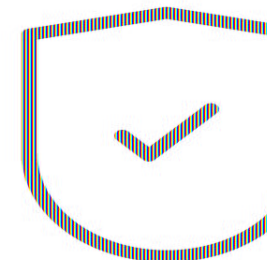
## Featured Projects



Insert project name



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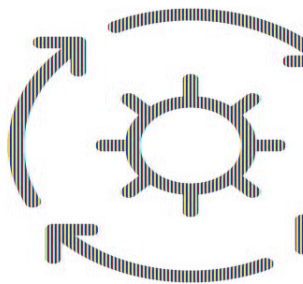
Insert project name

# Red Hat Research: Testing and ops

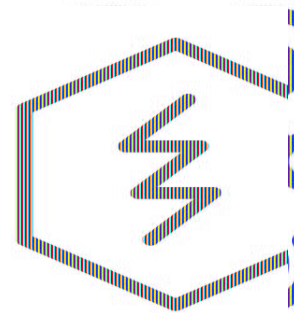


Great innovations don't amount to much if they can't be implemented and operated. That's the idea behind Red Hat's Office of the CTO's new motto: "Operate first!" Our researchers are contributing to new methods for systematizing and automating complex systems testing and then creating the frameworks and tools for operating them, often by embedding AI into predicting and avoiding failures or finding optimization pathways.

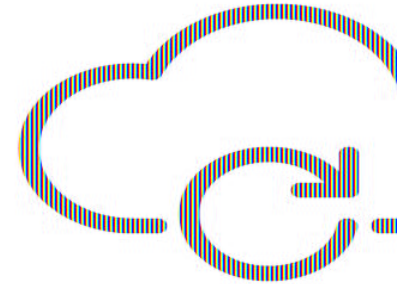
## Featured Projects



Insert project name



Insert project name



Insert project name

# Red Hat Research: AI for Cloud Ops

This project aims to address this gap in effective cloud management and operations with a concerted, systematic approach to building and integrating AI-driven software analytics into production systems. We aim to provide a rich selection of heavily-automated “ops” functionality as well as intuitive, easily-accessible analytics to users, developers, and administrators. In this way, our longer-term aim is to improve performance, resilience, and security in the cloud without incurring high operation costs.

## Project Repositories

- [Iter8 Online Experimentation Framework](#)
- [Praxi: Software Discovery using ML](#)
- [ACE: Approximate Concrete Execution](#)

## Project Team

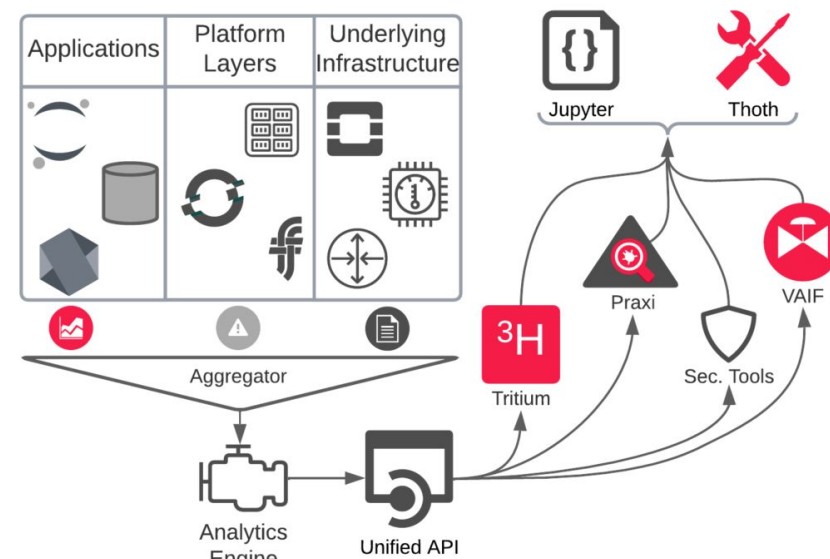
Principal Investigator: Ayse Coskun (BU)

Co-PIs: Alan Liu (BU) and Gianluca Stringhini (BU)

Red Hat Collaborators: Marcel Hild, Steven Huels, and Daniel Riek

IBM Collaborator: Fabio Oliveira

Graduate Students: Anthony Byrne, Mert Toslali, Saad Ullah, and Lesley Zhou



## Status

- Add a few points here as relevant
- 

## Opportunities

- Add a few points here as relevant

# Success Story Template

- Example of
  - A slide with
  - A fair amount of
  - Words
  -

# Partnerships: Driving results through strategic collaboration

University partners bring subject matter expertise, pathways to identify and develop talent, and influential stature in the open source community. Working with a variety of academics partners gives Red Hat the ability to pivot as technology trends change and new opportunities arise.

**Our growing network of partners includes**

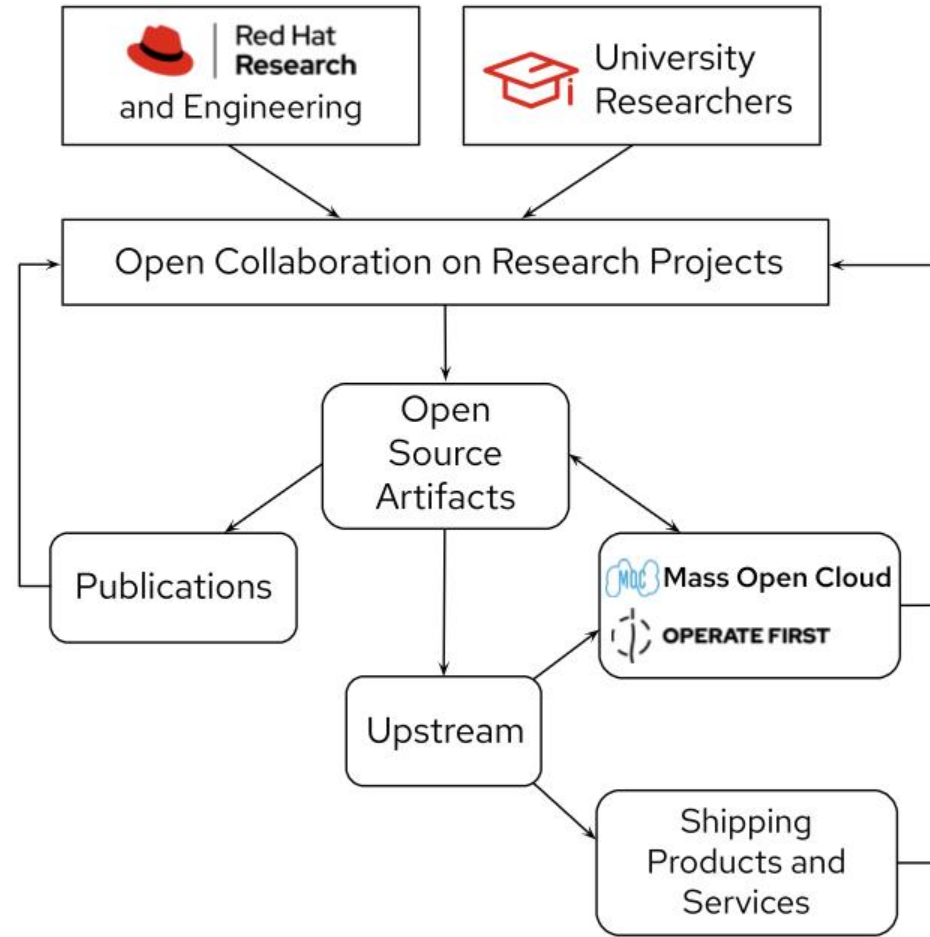


# Shared benefits of partnering to foster innovation in open source communities

- Train students to use the Red Hat stack - make OpenShift the default compute platform for academia
- Recruit new Red Hatters every year from university partners, all interested in scaling distributed systems for the hybrid cloud, to staff [cloud.redhat.com](https://cloud.redhat.com) as it grows
- Enable experiential learning opportunities that students are demanding while training them on our technologies.
- Take advantage of academic visibility in research in/on cloud to: accelerate open source, improved luster with federal agencies, medicine, new bleeding edge requirements.

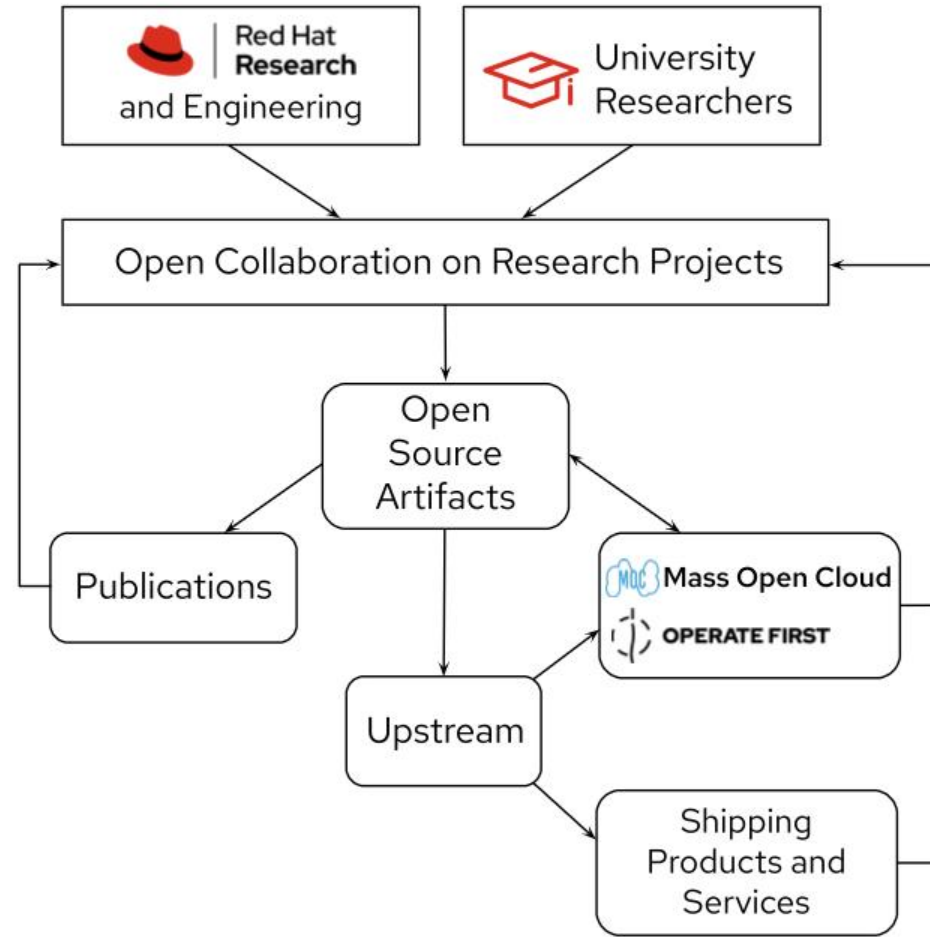
# A model to enable collaboration

Red Hat Research fosters collaborations between Red Hat engineers, universities, and partners to bring groundbreaking ideas into open source communities.



# Outputs from research collaborations amplify new discovery

- Open Source Artifacts
- Publications
- Upstream Contributions
- Infrastructure
- Product and Services

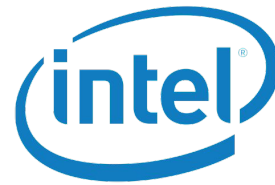


## The benefits of a global research program



### North America

Collaboratory, etc



### Israel

Notable things



### Europe

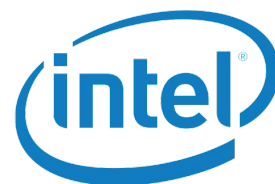
Thesis program, Horizon Europe, etc

## Engagement ? Building Community?



Research Days

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Research Interest Groups

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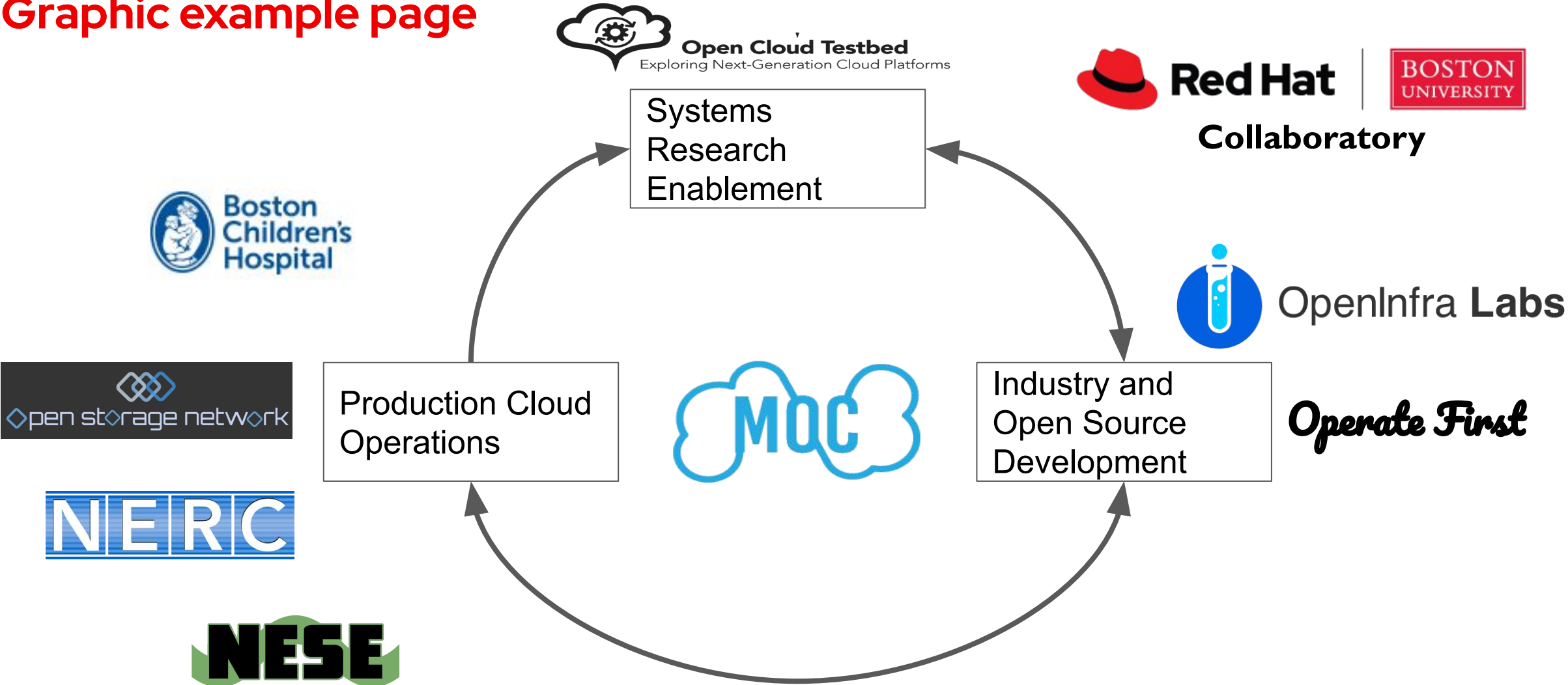
Red Hat Research Quarterly

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# Recap Call to Action

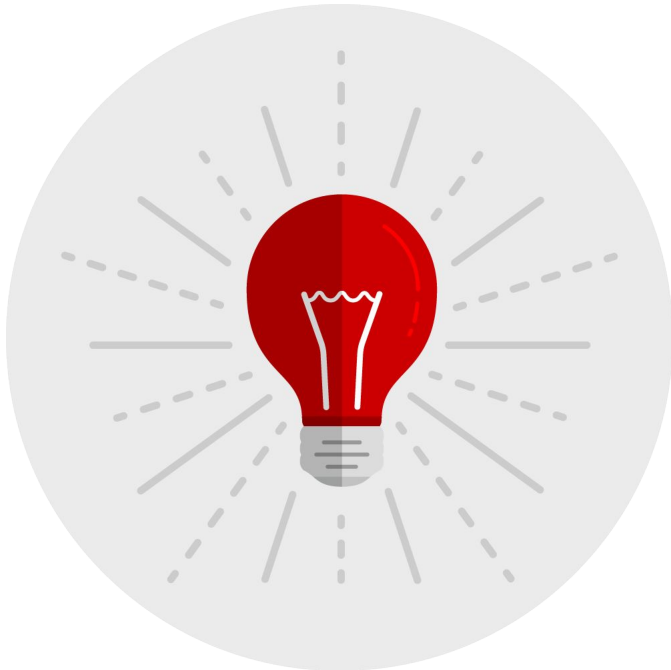


Graphic example page



Open Cloud Initiative (OCI)

## Data slide example



**XX**

Papers Published by Collaboratory Related Projects

**XX**

New Associated Grants Awarded

**XX**

Associated Grant Proposals in Progress

# LOTS OF TEXT PAGE

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  - A slide with
  - A fair amount of
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Join a Research Interest Group

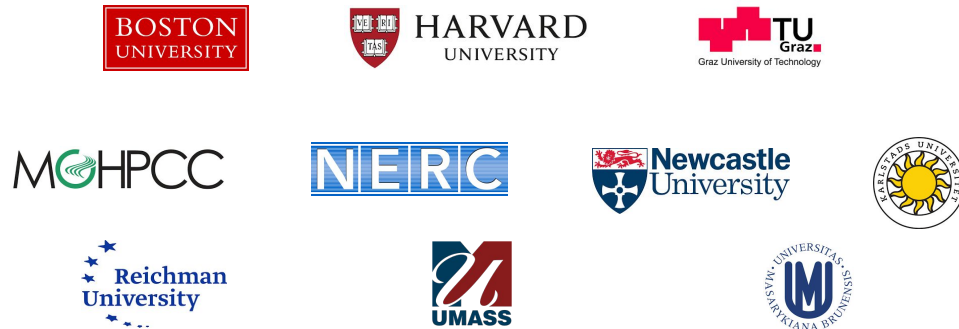
Learn more at [research.redhat.com/rigs](https://research.redhat.com/rigs)



# Red Hat Research: Leveraging research partnerships to propel emerging technologies

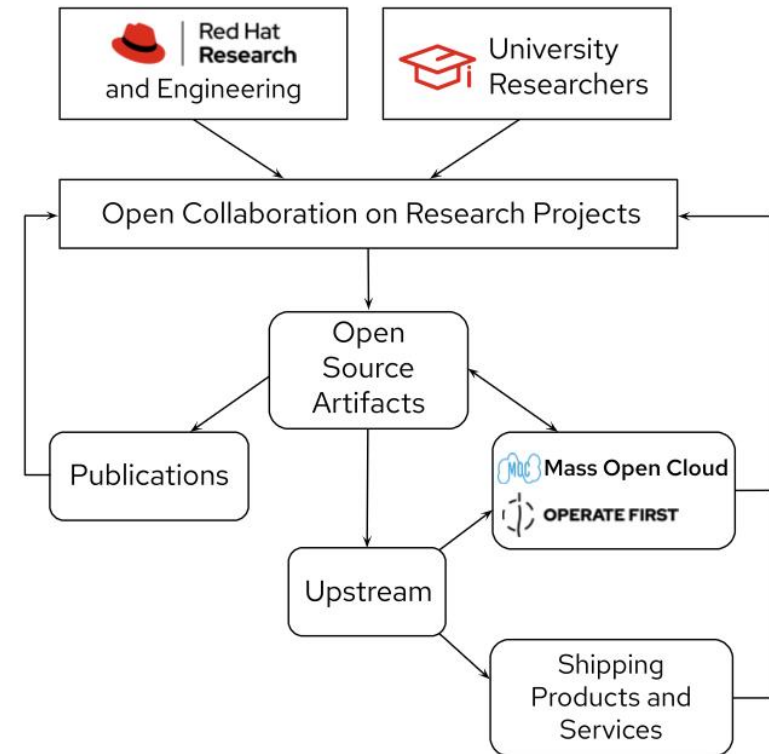
Red Hat Research fosters collaborations between Red Hat engineers, universities, and partners to bring groundbreaking ideas into open source communities.

## Our growing network of partners includes



## Recent global research investments

- AI for CloudOps (US)
- Data science for social sustainability (US/Europe)
- Energy efficient high performance computing (US)
- Programmable networking using BPF (Europe)
- Cloud cost optimization (Israel)



The virtuous cycle of open collaboration - Research success fuels open cloud growth